

270 marks

CANDIDATE DETAILS

**DAY and MONTH
of BIRTH** 

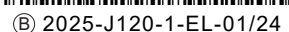
		/			For example, 3rd February is entered as 03/02														

TEACHER 

School Stamp

For Examiner only

Question	Mark
1.	
2.	
3.	
4.	
5.	
6.	
7.	
8.	
9.	
10.	
11.	
Total	
%	



Instructions

There are 11 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

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Question 1

(Suggested maximum time: 10 minutes)

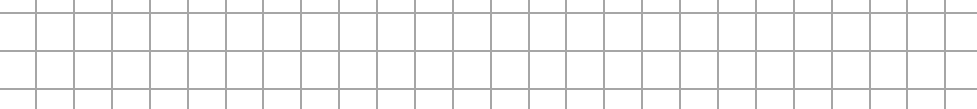
- (a) Millie has a savings jar of coins.
In the jar, she has a total of €14.10 made up of the following coins:
- 18 of 5 cent coins
 - 4 times as many 10 cent coins as 5 cent coins
 - The remaining coins are 20 cent coins



Work out how many 20 cent coins Millie has in her savings jar.



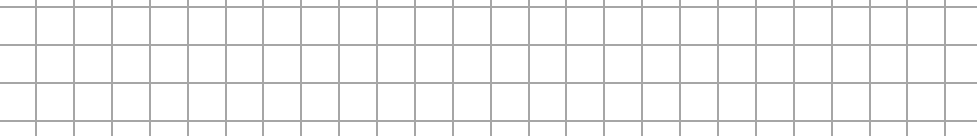
- (b)** 4 mechanics took 10 days to fix a jet engine.
Assuming they all work at the same speed,
work out how many days it would take 5 mechanics to fix the same engine.



- (c) In a clock shop a Grandfather clock chimes once every hour, a Mantel clock chimes once every 5 hours and a Pendulum clock chimes once every 90 minutes.



All 3 clocked chimed at 08:00 one morning.
Work out the next time they chimed together again.



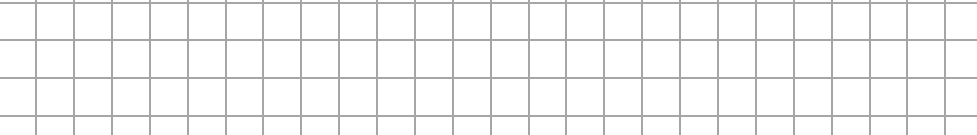
Question 2 (Suggested maximum time: 10 minutes)

Question 2 (Suggested maximum time: 10 minutes)

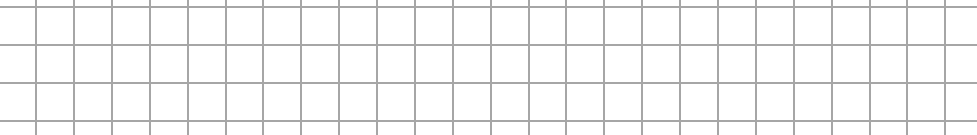
- (a)** Three of the first five terms of a linear sequence are shown below.

-14 , , 4 , , 22

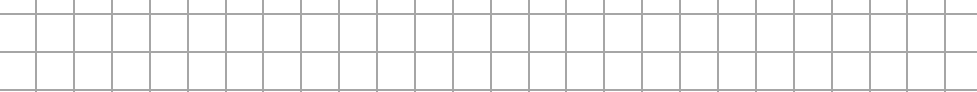
- (i) Fill in the missing terms to complete the first five terms of this sequence.



- (ii)** Show that the n th term of this sequence is $9n - 23$.



- (iii)** Hence, or otherwise, work out the **50th term** of this sequence.



(b) The first five terms of another sequence are as follows:

6 , 7 , 10 , 15 , 22

(i) Tick (✓) the correct box below to show what type of sequence this is.
Tick **one** box only. Justify your answer.

linear

1

quadratic

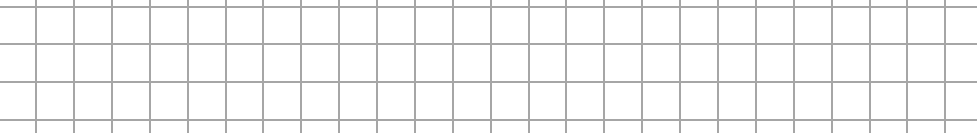
1

exponential

5

- Justification:

(ii) Write down the next three terms of this sequence.

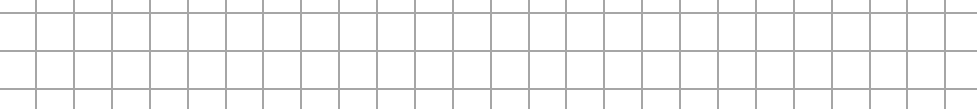


Question 3 (Suggested maximum time: 10 minutes)

A park hosted a 5 km park run.

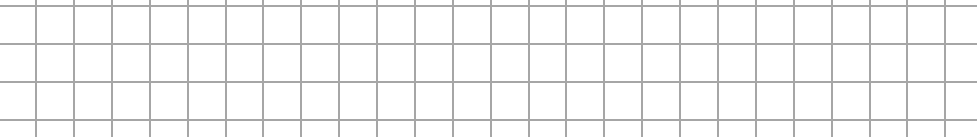
parkrun

[Note: 28 – 30 means 28 m or more, but less than 30 m, etc.]

- 

-

- (c) Work out the percentage of runners who completed the race in less than 30 minutes. Give your answer correct to the nearest whole number.



- (d) Use mid-interval values to estimate the **mean** time taken by the runners to finish the race. Give your answer correct to 1 decimal place.

[illegible]

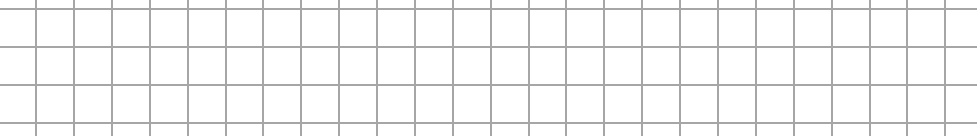
- (f) Work out the maximum range and the minimum range of times taken to complete the run. Give your answers correct to the nearest minute.

maximum range =

mins

minimum range =

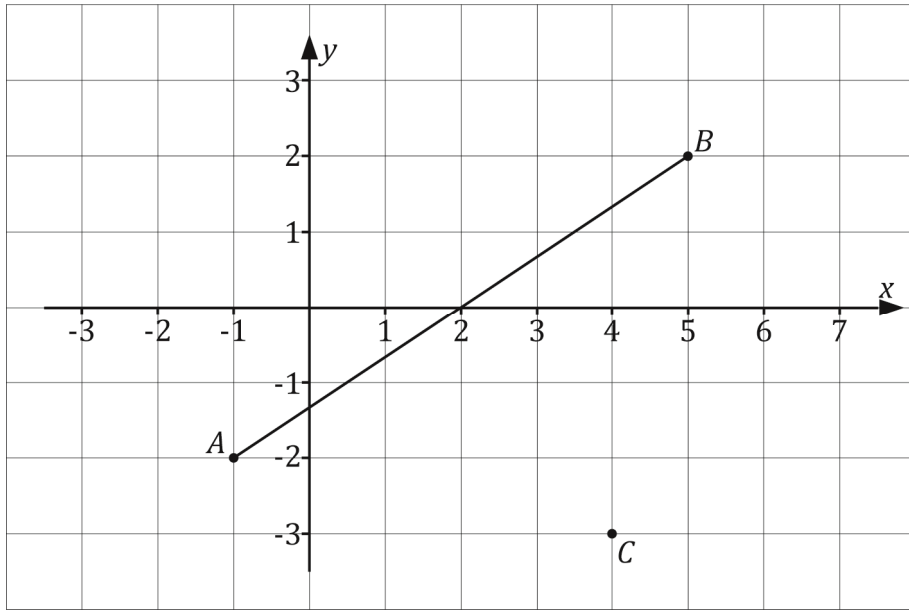
mins



Question 4 (Suggested maximum time: 10 minutes)

Question 4 (Suggested maximum time: 10 minutes)

The points A , B and C shown on the co-ordinate diagram below.



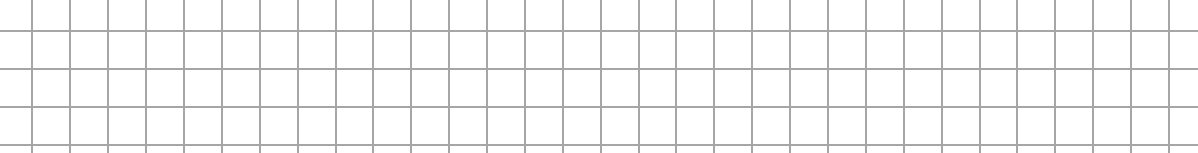
- (a)** Write the co-ordinates of the points A , B and C in the spaces below.

$$A = \begin{pmatrix} & \\ & \end{pmatrix} \quad B = \begin{pmatrix} & \\ & \end{pmatrix} \quad C = \begin{pmatrix} & \\ & \end{pmatrix}$$

- (b)** Write down the co-ordinates of the point M , the midpoint of $[AB]$.

$M = \left(\begin{array}{cc} & \end{array} \right)$

- (c)** Work out the slope of the line AB .



- (d) The equations of three lines are shown below.
Tick (✓) the correct box below to show which line is **perpendicular** to the line AB .
Tick **one** box only. Justify your answer.

$$y = 3x + 2$$

1

$$3y = 2x - 4$$

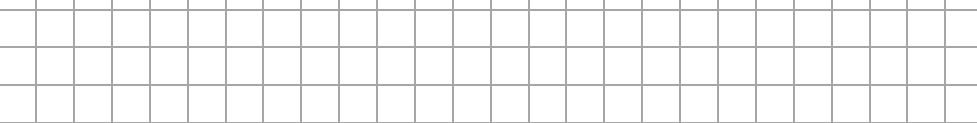
1

$$2y = -3x + 6$$

1

- Justification:

- (e)** Using the co-ordinate diagram, or otherwise, find the image of the point C under axial symmetry in the line AB .



Question 5**(Suggested maximum time: 10 minutes)**

- (a) Multiply out and simplify $4(2x + 5) - 3(x - 2)$.

- (b) Use your answer from (a) or otherwise to solve the equation

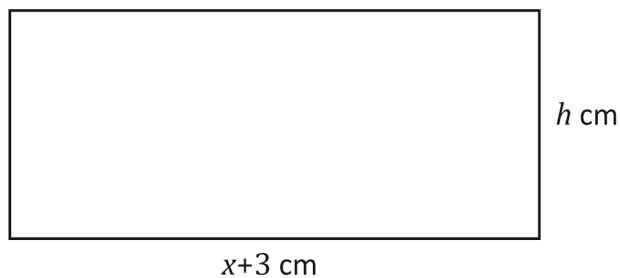
$$4(2x + 5) - 3(x - 2) = 101.$$

- (c) Factorise $49x^2 - 121$.

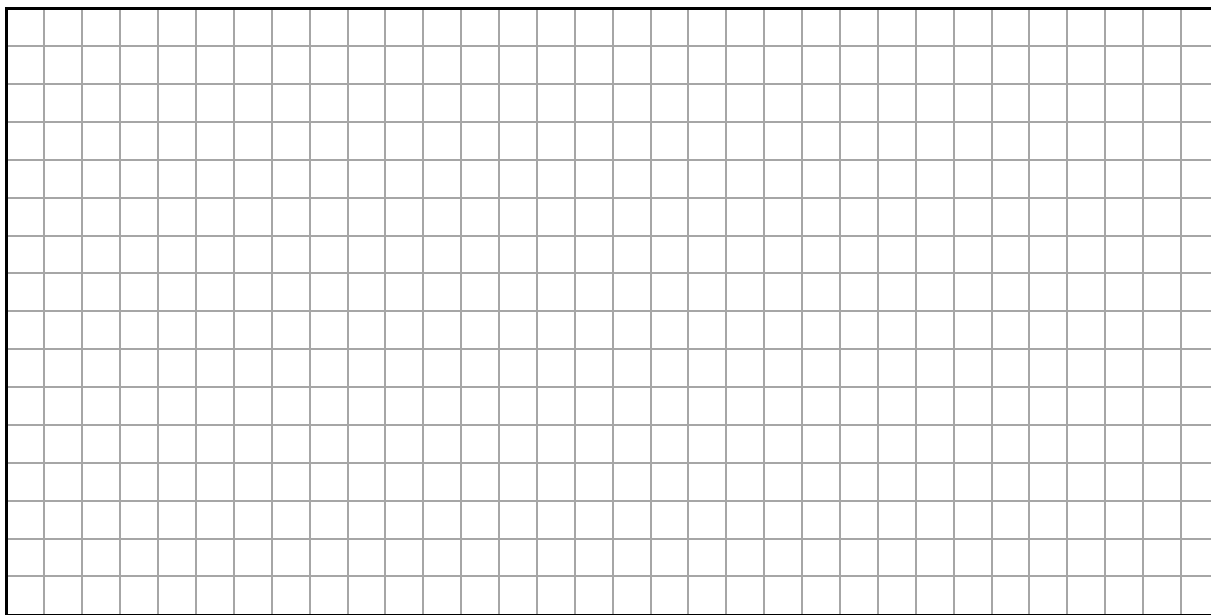
- (d) Factorise $4ac + 2a - 2cb - b$.



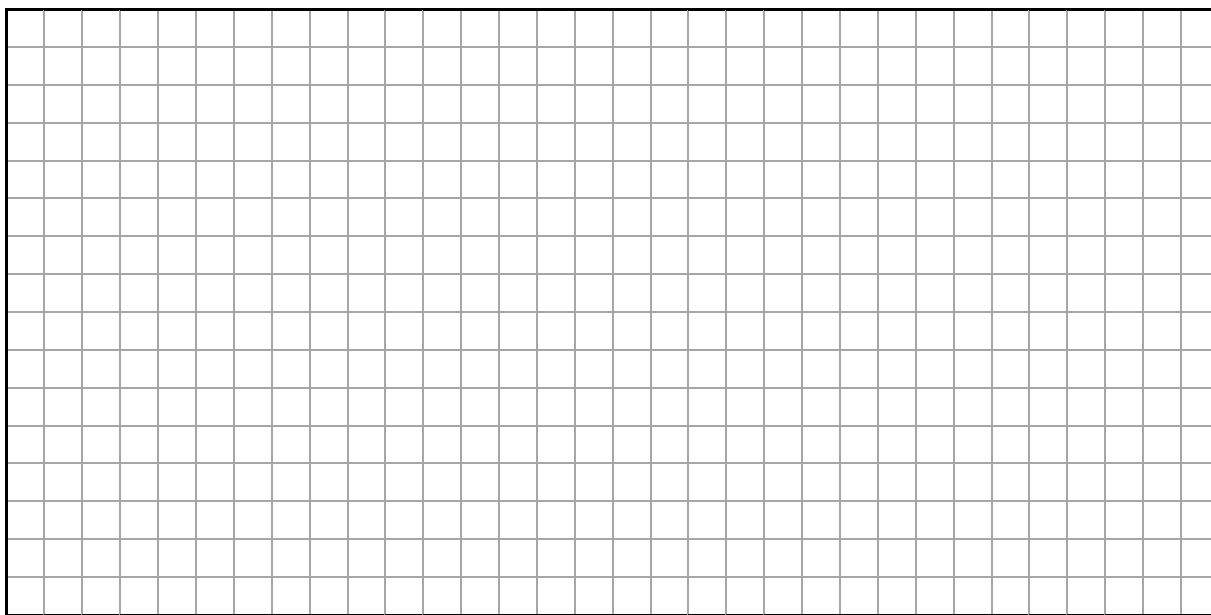
- (e) The length of a rectangle is $x + 3$ cm and its width is h cm as shown.



- (i) The area of the rectangle in terms of x is $x^2 - 2x - 15$.
By factorising $x^2 - 2x - 15$ or otherwise, find the value of h .



- (ii) Work out the value of x when the area of the rectangle is 84 cm^2 .

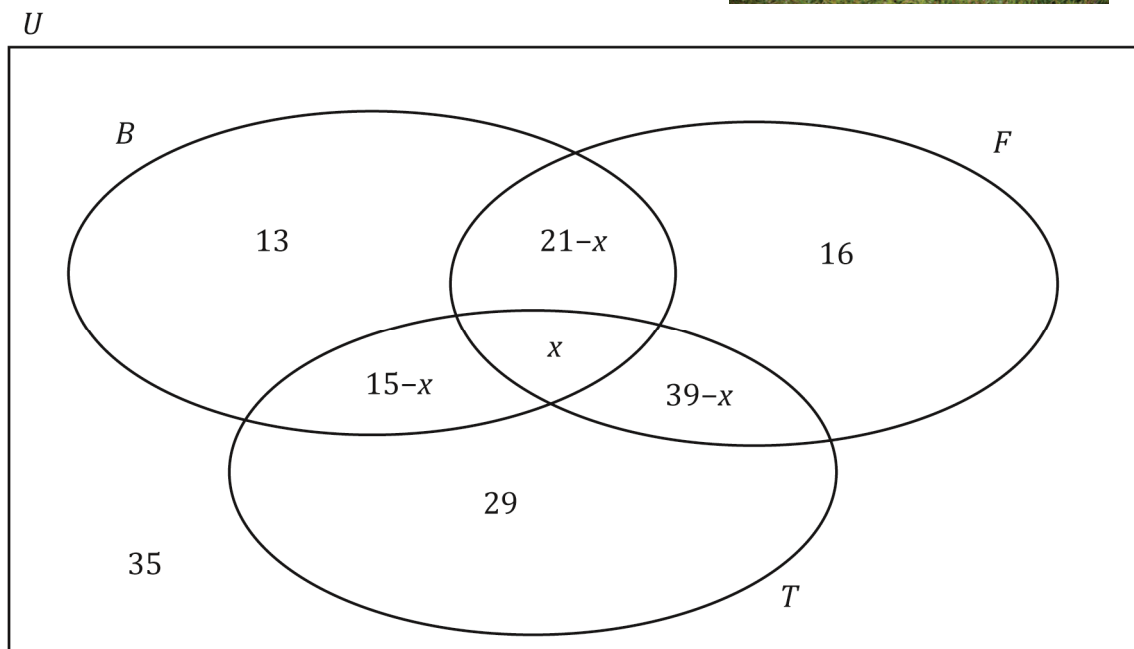


Question 6

(Suggested maximum time: 10 minutes)

A group of 150 students were asked which of the following sports they enjoyed playing: basketball (B), football (F) and tennis (T).

The responses from the students are shown in the Venn diagram below.



- (a) Tick (✓) the correct box to show what type of data is recorded above.
Tick **one** box only. Give a reason for your answer.

Categorical

Numerical

☐
☐

Reason:

- (b) Write down $\#(B \cap T)$.

Answer =

- (c) Write down $\#(F \setminus (B \cup T))$. Answer =



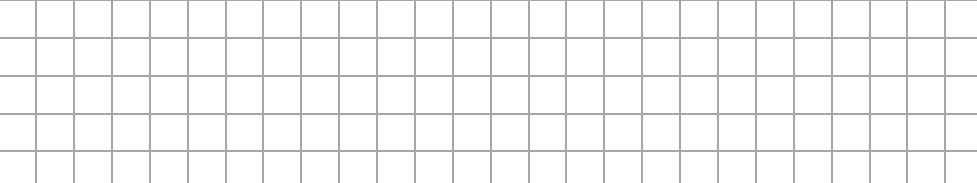
- (d)** Work out the number of students who enjoyed playing **only one** of the sports listed.

[illegible]

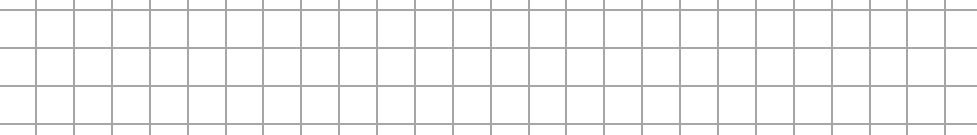
- (e) Work out the probability that a student chosen at random from the group did not enjoy playing any of the sports listed.



- (f)** Work out the value of x in the Venn diagram.



- (g) Work out the number of students who enjoyed playing **at least two** of the sports listed.

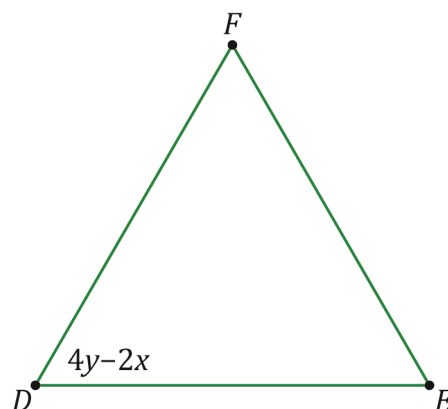
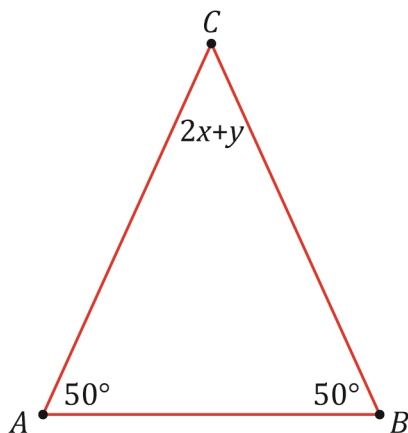


Question 7

(Suggested maximum time: 10 minutes)

The triangle ABC and the triangle DEF are shown below.

The angles $\angle BCA$ and $\angle EDF$ are given in terms of x and y , where $x, y \in \mathbb{Z}$.



- (a) Tick (✓) the correct box below to show what type of triangle ABC is.

Tick **one** box only. Justify your answer.

scalene

☐

right-angled

☐

isosceles

☐

Justification:

- (b) Work out the measure of the angle $|\angle BCA|$,
and hence write down an equation in x and y .

Equation in x and y :



- (c) The triangle DEF is an equilateral triangle.
 Tick (✓) the correct box below to show which of the following equations is true for x and y .
 Tick **one** box only. Justify your answer.

$$2y - x = 30$$

☐

$$2y - x = 90$$

☐

$$x + 2y = 60$$

☐

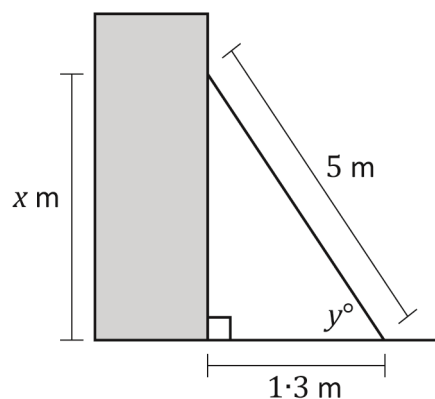
Justification:

- (d) Solve your equations from (b) and (c) to find the value of x and the value of y .

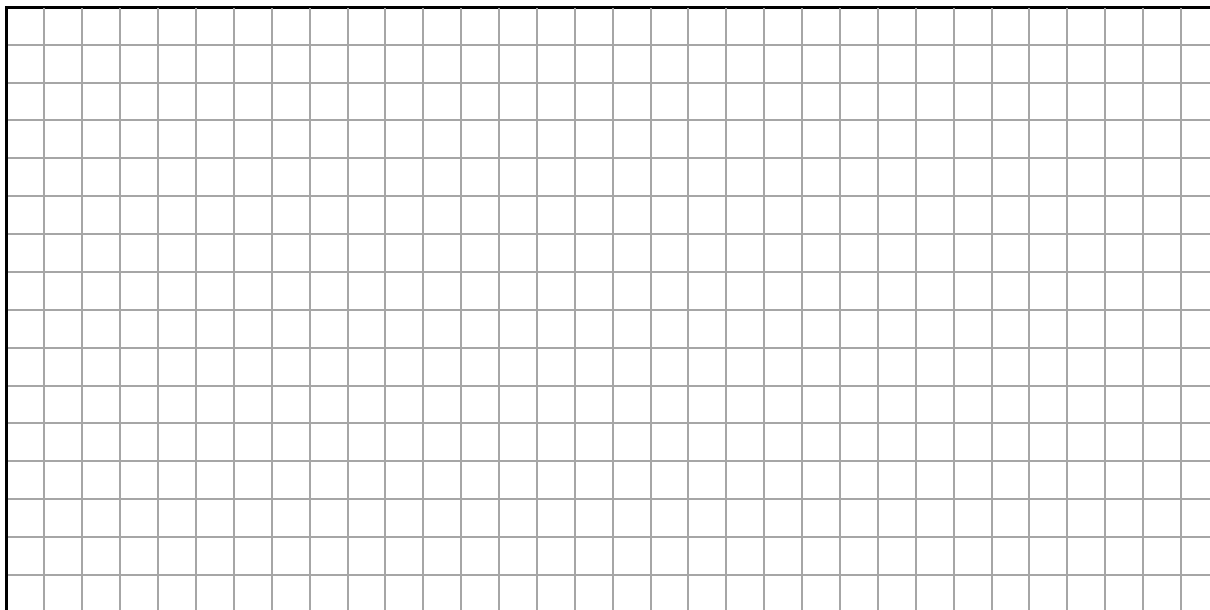


Question 8**(Suggested maximum time: 10 minutes)**

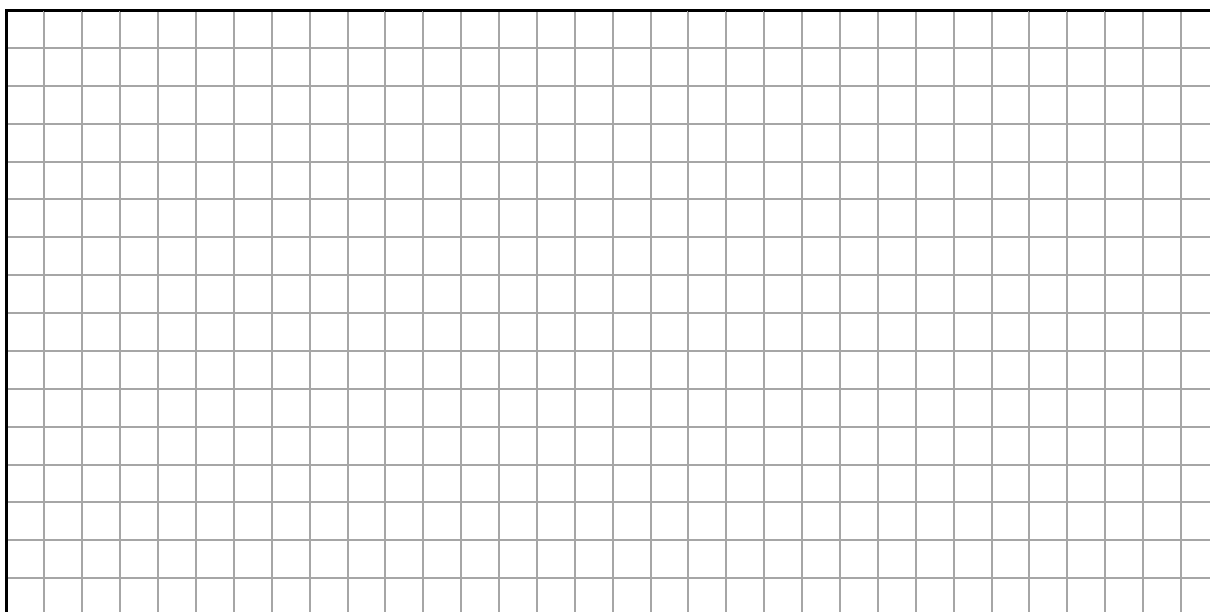
A ladder, 5 m long stands on level ground and reaches a point at a height of x m on a vertical wall. The foot of the ladder is 1.3 m from the base of the wall as shown in the diagram (not to scale).



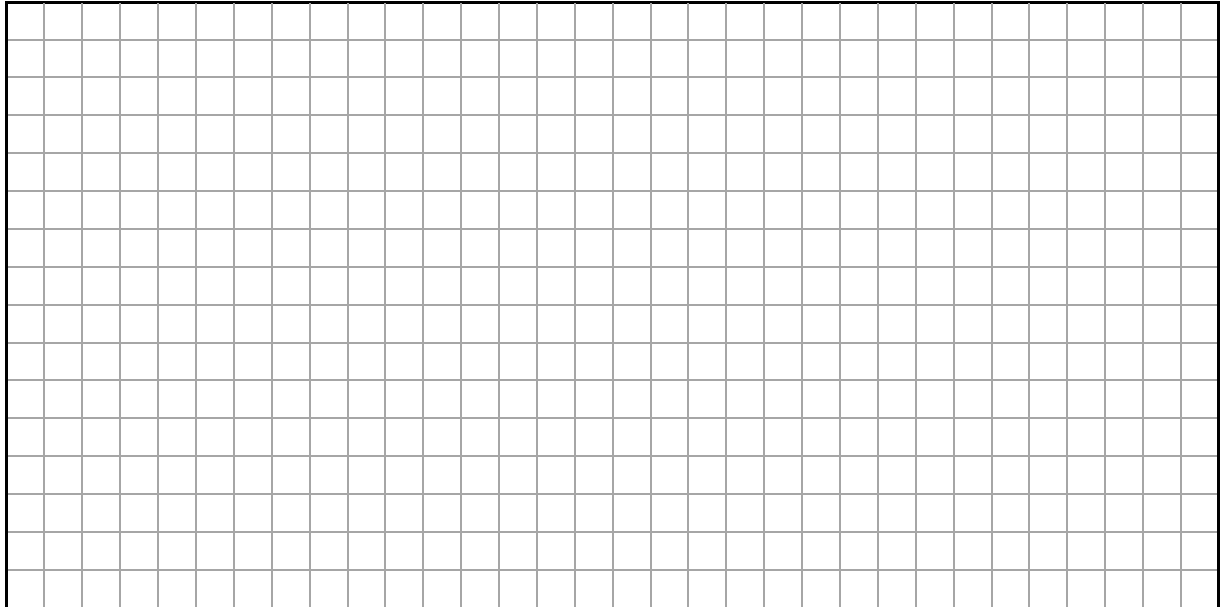
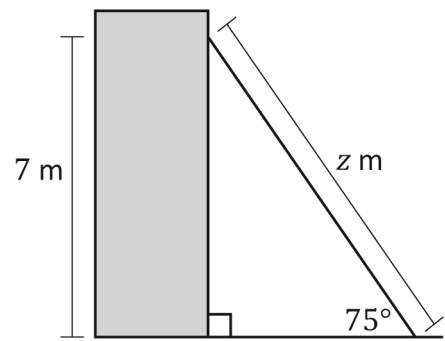
- (a) Work out the value of x .
Give your answer in metres correct to 1 decimal place.



- (b) Building safety regulations recommend that a ladder should make an angle of elevation of approximately 75° with level ground.
Show that this ladder is safe by finding y° , the angle of elevation of the ladder.



- (c) On another building, a longer ladder is needed to reach a point on a vertical wall 7 m from level ground. Work out z , the length of the ladder needed to reach this point. Give your answer in metres, correct to 2 decimal places.



Question 9 (Suggested maximum time: 15 minutes)

Question 9 (Suggested maximum time: 15 minutes)

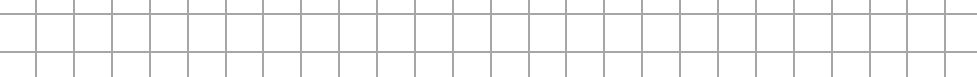
Brian works as a waiter for 30 hours a week and is paid a basic wage of €393.

- (a)** Work out Brian's hourly rate.

- (b)** Brian also earns tips each week.

On a particular week he earned a total of €500 including tips.

Express Brian's earnings in **tips** as a **percentage** of his total earnings that week.



- (c) Brian's gross income for last year was €28 000.

He pays tax at a rate of 20%.

His annual Tax Credit is €3995.

Work out Brian's **income tax** paid last year.

[illegible]

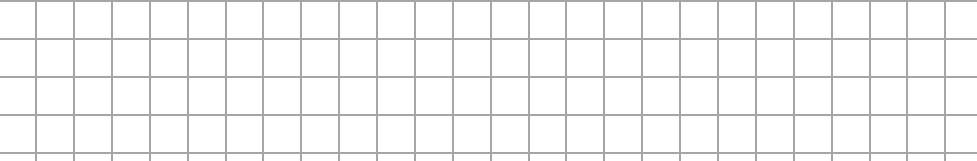
- (d)** Brian must pay Universal Social Charge (USC) on all his gross income. The income bands and rates of USC that Brian must pay are given in the table below. Brian pays USC at all three rates listed below.

Income Band	Rate of USC	USC paid by Brian
Up to €12 012		€60.06
From €12 012.01 to €27 382.00	2%	€307.40
From €27 382.01 to €70 044.00	4%	

Fill in the missing details in the table above.

[illegible]

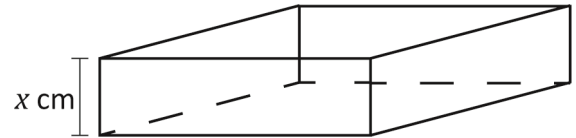
- (e) Last year, Brian also paid a total of €1129.34 in PRSI contributions.
Work out Brian's **net take home pay** for last year after tax, USC and PRSI have been paid.



Question 10

(Suggested maximum time: 5 minutes)

A rectangular box has a height of x cm as shown.



- (a) The measure of the width of the rectangle is twice the measure of the height and the measure of the length is twice its width.
Write down the width and the length of the box in terms of x .

Width of box =

cm

Length of box =

cm

[illegible]

- (b)** Show that the box has a total surface area of $28x^2 \text{ cm}^2$.

[illegible]

- (c)** The surface area of the box is 567 cm^2 . Work out the value of x .

A large grid of graph paper, consisting of 20 columns and 10 rows of squares, intended for drawing a picture.

- (d)** Find the volume of the box.

[illegible]

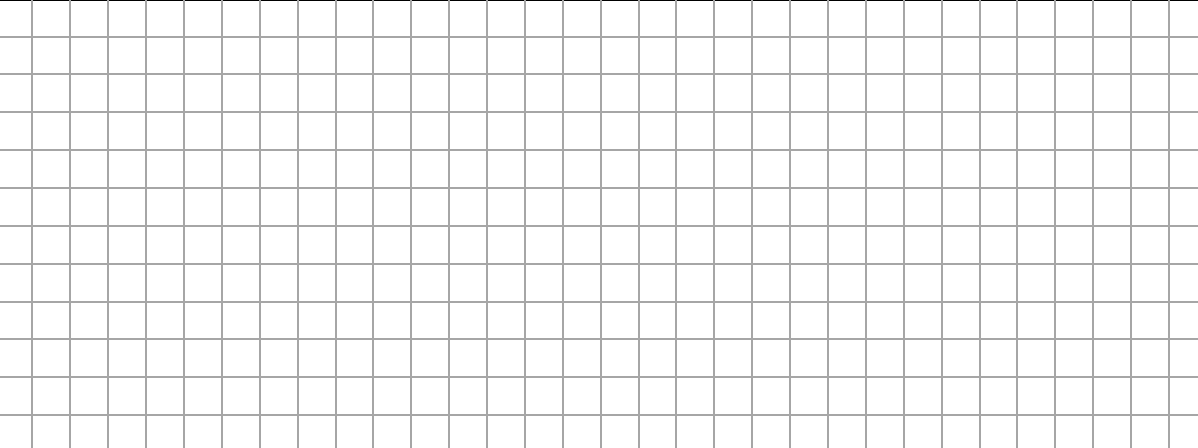
Question 11

(Suggested maximum time: 10 minutes)

The 24 Hours of Le Mans is an endurance-focused sports car race. The race is won by the car that covers the greatest distance in 24 hours.



- (a) The current record for the greatest distance covered during this 24 hour race is 5411 km.
- Work out the average **speed** of the record holder during this race in km per hour.
- Give your answer correct to 1 decimal place.
- 

A large grid of graph paper, consisting of 20 columns and 15 rows of squares, intended for drawing a picture.

- (b)** The car that set this record has a wheel with a diameter of 72 cm. Work out how many full rotations the wheel did over the distance of 5411 km. Give your answer, correct to two significant figures, in the form $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{N}$.

A full-page sheet of white graph paper with a light gray grid. The grid consists of small squares, approximately 10 units wide by 10 units high. There are no margins or additional markings on the page.

Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

Do not write on this page

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Pre-Junior Cycle Final Examination, 2025

Mathematics – Higher Level

Time: 2 hours

